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(57) Abstract:

An apparatus 200 for grinding food materials, the apparatus 200 includes a grinding stone assembly including a stationary grinding base 202 to keep grinding food materials and a grinding roller 204 to grind the food materials; and a moving mechanism comprising a flap linkage 206, a roller linkage 208, an electric motor 210, and a plurality of sensors 212 to control grinding operations such that the grinding roller 204 is rotated over the grinding base 202 including food materials, and the rotational motion of the grinding roller 204 converts into reciprocal motion to cover complete length of the grinding base 202 to grind the food materials in controlled speed of the grinding roller 204 to preserve texture and retain authentic taste of the food products.

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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The Patent Rules, 2003

COMPLETE SPECIFICATION

(See sections 10 & rule 13)

1. TITLE OF THE INVENTION

AUTOMATED BI-DIRECTIONAL GRINDING STONE APPARATUS

2. APPLICANT (S)

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3. PREAMBLE TO THE DESCRIPTION

COMPLETE SPECIFICATION

The following specification particularly describes the invention and the manner in which it is to be performed.

TECHNICAL FIELD

[0001] The present disclosure relates generally to the technical field of grinders. In particular, it pertains to a stone-based mechanically operated reciprocating grinding apparatus converting the continuous rotary motion of an electric motor into a controlled reciprocating motion for stone grinding process.

BACKGROUND

[0002] Background description includes information that may be useful in understanding the present disclosure. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed disclosure, or that any publication specifically or implicitly referenced is prior art.

[0003] Traditionally, food and spices grinding is normally done using stone grinders known as Sil-Batta that is manually operated through reciprocating action to grind food ingredients. The mechanism requires a continuous reciprocating action, thereby requiring considerable physical effort on the part of the user. The mortar and pestle are also used to manually crush and grind substances. One such prior art is depicted in FIG. 1, where a flat stone base 102 with a cylindrical stone 104 are used to grind spices 106. To grind the food material like spices and chatni, the food articles 106 are kept of the stone base 102 and manually moving the cylindrical stone 104 over the food articles 106 rotably and from lower end to upper end of the stone base, they are crushed and turns into fine powder. Water is added in between the process for wet products like chatni.

[0004] The process of grinding is manually through reciprocating action to grind food ingredients. The mechanism requires a continuous reciprocating action, thereby requiring considerable physical effort on the part of the user and time for the grinding operation, thus resulting in physical fatigue for the user.

[0005] The new appliances include electric mixer grinder and wet grinder which make use of rotating blades or grinding stones to facilitate quick food preparation through motor operation. These appliances either make use of manual operation or rotating blades/stone operation but they do not automate the reciprocating action of the traditional stone grinder. Such appliances operate at

very high speeds, thereby generating heat that affects the texture, smell and nutritional values of food items and not being able to achieve the required food quality. Food processing machines with rotating mechanisms are large, expensive and not suitable for small-scale operations or for domestic use.

[0006] Recent advancements in the field have focused on improving the stone grinder by addressing above issues. One such Patent Document US10421077B2 discloses a large capacity upright tilting wet stone grinder of capacity 125 to 300 litres comprising a cylindrical vessel; a centre axle attached to and positioned vertically in the centre of the vessel; an adjustable speed gear driven motor positioned beneath the vessel connected to rotate the centre axle; granite grinding stone positioned on opposite sides of the centre axle for grinding against the vessel's inside bottom; and a tilt frame connected to top of the centre axle and the motor tilted using a gear drive and upright position.

[0007] While cited references is based on rotational grinding within a drum and does not involve reciprocating or oscillatory motion, there is a possibility to provide better solution that can address the abovementioned problem.

[0008] There is, therefore, a need to provide a small, energy-efficient and user-friendly automated apparatus that works on the concept of the traditional stone grinding method to address abovementioned drawbacks.

OBJECTS OF THE DISCLOSURE

[0009] A general object of the present disclosure is to provide an apparatus that modernize the traditional grinding stone process by introducing mechanical automation while keeping the grinding characteristics of stone-based food processing intact.

[0010] An object of the present disclosure is to provide an apparatus for solution that retains the authentic taste and feel of traditionally ground food products while ensuring increased ease of use in food processing environments.

[0011] Another object of the present disclosure is to provide a compact, energy-efficient and user-friendly apparatus that works on the concept of the traditional stone grinding.

[0012] Another object of the present disclosure is to provide a sensor-based feedback mechanism that continuously monitors the load, motor current and position of motion to evaluate the grinding condition while maintaining the traditional grinding effects.

[0013] Another object of the present disclosure is to provide the grinding stone that functions to reciprocate motion of the grinding stone over stone base, for efficient grinding.

SUMMARY

[0014] Aspects of the present disclosure relate to the technical field of grinders. In particular, it pertains to a stone-based mechanically operated reciprocating grinding apparatus converting the continuous rotary motion of an electric motor into a controlled reciprocating motion for stone grinding process. More particular, the proposed disclosure operates on the principle of converting the continuous rotary motion of an electric motor into a controlled reciprocating motion combined with partial rotational motion, thereby replicating and enhancing the traditional flat stone grinding process.

[0015] In an aspect, the disclosure is about an apparatus for grinding food materials, the apparatus includes a grinding stone assembly including a stationary grinding base to keep grinding food materials and a grinding roller to grind the food materials; and a moving mechanism including a flap linkage, a roller linkage, an electric motor, and a plurality of sensors to control grinding operations such that the grinding roller is rotated over the grinding base including food materials, and the rotational motion of the grinding roller converts into reciprocal motion to cover complete length of the grinding base to grind the food materials in controlled speed of the grinding roller to preserve texture and retain authentic taste of the food products.

[0016] In an embodiment, the apparatus is compact, energy-efficient and user-friendly grinding solution that retains the authentic taste and feel of traditionally ground food products.

[0017] In an embodiment, the electric motor drives the roller linkage to grind the food materials.

[0018] In an embodiment, the roller linkage includes a crank and link mechanism and an actuating linkage arm.

[0019] In an embodiment, the crank and link mechanism converts the rotary motion of the grinding stone to reciprocating motion.

[0020] In an embodiment, the plurality of sensors includes a load cell sensor, a motor current sensor, and a position sensor to monitor grinding resistance, motor load and positional movement respectively, enabling automatic control of grinding and grinded material discharge process based on feedback mechanism.

[0021] In an embodiment, the conversion of rotary motion into a reciprocating motion having two degrees of freedom (2-DOF) at the grinding stone including linear reciprocating motion along the surface of the grinding base, and angular oscillatory motion, and the grinder stone rotates partially from approximately 0° to 180° and returns to 0° instead of undergoing full rotation.

[0022] In an embodiment, the apparatus automatically detects completion of the grinding cycle once predefined threshold values are reached indicated by the decrease resistance and motor load.

[0023] In an embodiment, a flap mechanism is activated for discharge and collection of the grinded food materials when the material is grinded as per requirement.

[0024] Another aspect of the disclosure is a method for grinding food materials using a grinding apparatus including steps for placing of food martials over a stationary grinding base of the grinding apparatus; starting an electric motor to rotate a grinding stone to have reciprocating motion over the food materials; generating compressive and shear force for grinding stone to grind the food materials; controlling speed of the grinding stone automatically through sensor-based feedback; ending grinding automatically once the sensed parameters of grinded food materials reaches predefined threshold values indicating completion of grinding; and actuating discharge flap to discharge the grinded food materials for use and stopping motor.

[0025] Various objects, features, aspects and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like components.

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The accompanying drawings are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

[0027] FIG. 1 illustrates a traditional stone grinder sil-batta that is manually operated through reciprocating action to grind food ingredients.

[0028] FIG. 2 illustrates an exemplary view of stone-based mechanically operated reciprocating grinding apparatus, in accordance with embodiments of the present disclosure.

[0029] FIGs. 3A, 3B, and 3C illustrate different sub-assemblies of the stone-based grinding apparatus of FIG. 2, in accordance with embodiments of the present disclosure.

[0030] FIG. 4 illustrates an exemplary method block diagram for operating stone-based mechanically operated reciprocating grinding apparatus, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0031] The following is a detailed description of embodiments of the disclosure depicted in the accompanying drawings. The embodiments are in such details as to clearly communicate the disclosure. However, the amount of detail offered is not intended to limit the anticipated variations of embodiments; on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

[0032] Embodiments explained herein relate to the technical field of grinders. In particular, it pertains to a stone-based mechanically operated reciprocating grinding apparatus converting the continuous rotary motion of an electric motor into a controlled reciprocating motion for stone grinding process.

[0033] In an embodiment, the proposed disclosure is about an apparatus for grinding food materials, the apparatus includes a grinding stone assembly including a stationary grinding base to keep grinding food materials and a grinding roller to grind the food materials; and a moving mechanism comprising a flap linkage, a roller linkage, an electric motor, and a plurality of sensors to control grinding operations such that the grinding roller is rotated over the grinding base including food materials, and the rotational motion of the grinding roller converts into reciprocal motion to cover complete length of the grinding base to grind the food materials in controlled speed of the grinding roller to preserve texture and retain authentic taste of the food products.

[0034] In an embodiment, the apparatus is compact, energy-efficient and user-friendly grinding solution that retains the authentic taste and feel of traditionally ground food products.

[0035] In an embodiment, the electric motor drives the roller linkage to grind the food materials.

[0036] In an embodiment, the roller linkage includes a crank and link mechanism and an actuating linkage arm.

[0037] In an embodiment, the crank and link mechanism converts the rotary motion of the grinding stone to reciprocating motion.

[0038] In an embodiment, the plurality of sensors includes a load cell sensor, a motor current sensor, and a position sensor to monitor grinding resistance, motor load and positional movement respectively, enabling automatic control of grinding and grinded material discharge process based on feedback mechanism.

[0039] In an embodiment, the conversion of rotary motion into a reciprocating motion having two degrees of freedom (2-DOF) at the grinding stone including linear reciprocating motion along the surface of the grinding base,

and angular oscillatory motion, and the grinder stone rotates partially from approximately 0° to 180° and returns to 0° instead of undergoing full rotation.

[0040] In an embodiment, the apparatus automatically detects completion of the grinding cycle once predefined threshold values are reached indicated by the decrease resistance and motor load.

[0041] In an embodiment, a flap mechanism is activated for discharge and collection of the grinded food materials when the material is grinded as per requirement.

[0042] Referring now to FIG. 2 where an exemplary view of stone-based mechanically operated reciprocating grinding apparatus 200 is shown. The proposed apparatus operates on the principle of converting the continuous rotary motion of a motor into a controlled reciprocating motion combined with partial rotational motion, thereby replicating and enhancing the traditional flat stone grinding process. The raw material, such as grains, spices or leafy items, is placed on the flat stationary grinding base to convert them into powder form or chatni.

[0043] In an embodiment, the apparatus 200 for grinding food materials, the apparatus 200 includes a grinding stone assembly including a stationary grinding base 202, a grinding roller 204; and a moving mechanism including a flap linkage 206, a roller linkage 208, an electric motor 210, and a plurality of sensors 212.

[0044] In an embodiment, the stationary grinding base 202 configured with a grinding basin used to place food materials for grinding. The grinding roller 204, which is cylindrical in shape, is used to grind the food materials are moved using the moving mechanism.

[0045] In an embodiment, the moving mechanism includes a flap linkage 206, a roller linkage 208, and an electric motor 210.

[0046] In an embodiment, the plurality of sensors 212 controls the grinding operations. The sensors 212 includes a load cell sensor 212-A to monitor grinding resistance, a motor current sensor 212-B to monitor motor load, and a position sensor 212-C (collectively referred a sensors 212) to sense positional movement of the grinding roller 204. The sensors 212 enabling automatic control of grinding and grinded material discharge process based on feedback control system.

[0047] In an embodiment, the sensor-based feedback control system enhances the intelligence and automation of the grinding process. This utilizes a load cell or force sensor 212-A, motor current sensor 212-B and position sensor 212-C, which is a limit switch, to continuously monitor the grinding condition. This closed-loop feedback ensures consistent grinding quality, optimized operation and reduced energy consumption which is not present in existing traditional or modern grinding arts.

[0048] In an embodiment, the automated grinding stone apparatus 200 is designed to replicate the traditional grinding process using a mechanical setup including the flap linkage 206, which is connected to the roller linkage 208. The flap linkage 206 mounts the position sensor 212-C to control the distance for back and forth motion of the grinding stone 204 moved by the roller linkage 208.

[0049] In an embodiment, the electric motor 210 is coupled with the roller linkage 208 and automatically controlled by the feedback provided by the load cell sensor 212-A. The electric motor 210 is further controlled by the current sensor 212-B and safeguard the motor 210 against the draw of high current in case load increases.

[0050] In an embodiment, the apparatus 200 includes an automated flap discharge mechanism to ensure proper material handling. The discharge mechanism includes a flap, which remains in a slightly elevated position, allowing unobstructed grinding. After grinding cycles, the flap lowers to contact the stone base 202 and moves to push the final grinding material toward the defined collection zone. Once this action is completed, the flap lifts back to its original position and returns without disturbing the material, enabling continuous operation.

[0051] In an embodiment, the proposed apparatus 200 operates on the principle of converting the continuous rotary motion of the electric motor 210 into a controlled reciprocating motion combined with partial rotational motion, thereby replicating and enhancing the traditional base stone grinding process.

[0052] Ion an embodiment, initially, the raw food material, such as grains, spices or leafy materials, is placed on the chamber of the stationary stone grinding

base. After preparation, when the motor 210 is switched on, the motor 210 produces continuous rotary motion which is transmitted through a crank and link mechanism or quick return mechanism 214. This mechanism 214 converts the rotary input into a combined motion having two degrees of freedom (2-DOF).

[0053] In an embodiment, the two degrees of freedom at the grinding stone 204 are first, linear reciprocating motion along the grinding surface and second, angular oscillatory motion. This results the grinding stone 204 rotates partially from approximately 0° to 180° and returns to 0°, instead of undergoing full rotation. This combined motion generates both compressive forces due to downward pressure and shear forces due to the relative motion between the grinding surfaces of the stone base 202 and the grinding roller 204, resulting in efficient size reduction and uniform mixing of the food material.

[0054] In an embodiment, the quick return mechanism 214 ensures that the forward working stroke occurs at a slower speed for effective grinding, while the return stroke is faster, thereby minimizing idle time and improving operational efficiency. The grinding stone 204 continuously travels from one end of the base 202 to the other and returns in a cyclic manner until uniform grinding is achieved.

[0055] In an embodiment, the sensor-based feedback enables real-time monitoring and control of the grinding process. As the material becomes finer, the resistance and motor load decrease which is detected by the load cell sensor 212-A. Based on predefined threshold values, the apparatus automatically determines the completion of grinding and accordingly controls the motor 210 operation and activates the discharge mechanism to push the final grinding material toward the defined collection zone.

[0056] In an embodiment, as the process continues, the material is progressively ground into a fine paste or flour with consistent texture and quality. The integration of two-degree-of-freedom motion, quick return mechanism, automated material handling and sensor-based feedback control results in improved efficiency, reduced manual effort and preservation of traditional grinding characteristics.

[0057] In an embodiment, the apparatus 200 is compact, energy-efficient and user-friendly grinding solution that retains the authentic taste and feel of traditionally ground food products.

[0058] FIGs. 3A, 3B, and 3C illustrate different sub-assemblies of the stone-based grinding apparatus 200.

[0059] In an embodiment, as the proposed apparatus 200 operates on the principle of converting the continuous rotary motion of an electric motor 210 into a controlled reciprocating motion combined with partial rotational motion, the electric motor 210 drives the roller linkage 208 to grind the food materials.

[0060] In an embodiment, the flap linkage 206 (FIG. 3A) includes a flap frame 302 coupled with a pair of arm 304-1 and 304-2 (collectively referred as arms 304, herein). The arms 304 include the position sensor 212-C, which is a limit switch. The limit switch 212-C sense to and forth movement, as shown by the arrow, of the roller using movement of the flap frame 302 moved by the roller linkage 208. When position sensor 212-C senses the pre-set extreme position of the flap frame 302 at the upper end of the stone base 202, the return of the stone roller 204 is ensured.

[0061] In an embodiment, the roller linkage 208 (FIG. 3B) includes a crank and link mechanism 214. The mechanism 214 includes a fixed arm coupled to the motor 210, a first arm 306, and a second arm 308 coupled with a supporting link frame 310. The frame 310 is coupled with the grinding stone 204 and attached to the stone base 202 through an actuating linkage arm 216.

[0062] In an embodiment, the crank and link mechanism 214 converts the rotary motion of the grinding stone 204 to reciprocating motion. The conversion of rotary motion into a reciprocating motion having two degrees of freedom (2-DOF) at the grinding stone 204 including linear reciprocating motion along the surface of the grinding base 202, and angular oscillatory motion, and the grinder stone 204 rotates partially from approximately 0° to 180° and returns to 0° instead of undergoing full rotation.

[0063] In an embodiment, the motor current sensor 212-B, and a position sensor 212-C to monitor grinding resistance, and positional movement

respectively, enabling automatic control of grinding and grinded material discharge process based on feedback mechanism.

[0064] In an embodiment, FIG. 3C depicts the stationary grinding base 202 configured with a grinding basin used to place food materials for grinding. The load cell sensor 212-A is coupled to the base 202 for real-time monitoring of resistance encountered by the load cell and current sensor 212-B. the flap discharge mechanism is incorporated to ensure proper material handling during discharge.

[0065] FIG. 4 illustrates an exemplary method 400 block diagram for operating stone-based mechanically operated reciprocating grinding apparatus 200. The method 400 for grinding food materials using a grinding apparatus 200 including step 402 for placing of food martials over a stationary grinding base 202 of the grinding apparatus 200.

[0066] In an embodiment, method step 404 defines for starting an electric motor 210 to rotate a grinding stone 204 to have reciprocating motion over the food materials.

[0067] In an embodiment, method step 406 defines for generating compressive and shears force for grinding stone 204 to grind the food materials.

[0068] In an embodiment, method step 408 defines for controlling speed of the grinding stone 204 automatically through sensor-based feedback.

[0069] In an embodiment, method step 410 defines for ending grinding automatically once the sensed parameters of grinded food materials reaches predefined threshold values indicating completion of grinding.

[0070] In an embodiment, method step 412 defines for and actuating discharge flap to discharge the grinded food materials for use and stopping motor 210.

[0071] Thus, proposed apparatus 200 is unique as it combines a quick return mechanism with a mechanically operated reciprocating grinding integrated into a conventional flat stone grinding process. This enables automation of the grinding operation while retaining the inherent advantages of traditional stone-based grinding. The disclosed grinding apparatus 200 enables efficient, uniform,

intelligent and automated grinding, while preserving the advantages of traditional stone grinding, significantly reducing manual effort and improving operational productivity.

[0072] While the foregoing describes various embodiments of the invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof. The scope of the invention is determined by the claims that follow. The invention is not limited to the described embodiments, versions or examples, which are included to enable a person having ordinary skill in the art to make and use the invention when combined with information and knowledge available to the person having ordinary skill in the art.

ADVANTAGES OF THE DISCLOSURE

[0073] The present disclosure provides an apparatus that modernize the traditional grinding stone process by introducing mechanical automation while keeping the grinding characteristics of stone-based food processing intact.

[0074] The present disclosure provides an apparatus for solution that retains the authentic taste and feel of traditionally ground food products while ensuring increased ease of use in food processing environments.

[0075] The present disclosure provides a compact, energy-efficient and user-friendly apparatus that works on the concept of the traditional stone grinding.

[0076] The present disclosure provides a sensor-based feedback mechanism that continuously monitors the load, motor current and position of motion to evaluate the grinding condition while maintaining the traditional grinding effects.

[0077] The present disclosure provides the grinding stone that functions to reciprocate motion of the grinding stone over stone base, for efficient grinding.

We Claim:

1. An apparatus (200) for grinding food materials, the apparatus (200) comprising:

a grinding stone assembly comprising a stationary grinding base (202) to keep grinding food materials and a grinding roller (204) to grind the food materials; and

a moving mechanism comprising a flap linkage (206), a roller linkage (208), an electric motor (210), and a plurality of sensors (212) to control grinding operations:

wherein the grinding roller (204) is rotated over the grinding base (202) comprising food materials, and wherein the rotational motion of the grinding roller (204) converts into reciprocal motion to cover complete length of the grinding base (202) to grind the food materials in controlled speed of the grinding roller (204) to preserve texture and retain authentic taste of the food products.

2. The apparatus as claimed in claim 1, wherein the apparatus (200) is compact, energy-efficient and user-friendly grinding solution that retains the authentic taste and feel of traditionally ground food products.

3. The apparatus as claimed in claim 1, wherein the electric motor (210) drives the roller linkage (208) to grind the food materials.

4. The apparatus as claimed in claim 1, wherein the roller linkage (208) comprising a crank and link mechanism (214) and an actuating linkage arm (216).

5. The apparatus as claimed in claim 4, wherein the crank and link mechanism (214) converts the rotary motion of the grinding stone (204) to reciprocating motion.

6. The apparatus as claimed in claim 1, wherein the plurality of sensors (212) comprises a load cell sensor (212-A), a motor current sensor (212-B), and a position sensor (212-C) to monitor grinding resistance, motor load and positional movement respectively, enabling automatic control of grinding and grinded material discharge process based on feedback mechanism.

7. The apparatus as claimed in claim 1, wherein the conversion of rotary motion into a reciprocating motion having two degrees of freedom (2-DOF) at the grinding stone (204) comprising linear reciprocating motion along the surface of the grinding base (202), and angular oscillatory motion, wherein the grinder stone (204) rotates partially from approximately 0° to 180° and returns to 0° instead of undergoing full rotation.

8. The apparatus as claimed in claim 1, wherein the apparatus (200) automatically detects completion of the grinding cycle once predefined threshold values are reached indicated by the decrease resistance and motor load.

9. The apparatus as claimed in claim 1, wherein a flap mechanism is activated for discharge and collection of the grinded food materials when the material is grinded as per requirement.

10. A method (400) for grinding food materials using a grinding apparatus (200), the method (400) comprising steps for:

placing of food materials over a stationary grinding base (202) of the grinding apparatus (200);

starting an electric motor (210) to rotate a grinding stone (204) to have reciprocating motion over the food materials;

generating compressive and shear force for grinding stone (204) to grind the food materials;

controlling speed of the grinding stone (204) automatically through sensor-based feedback;

ending grinding automatically once the sensed parameters of grinded food materials reaches predefined threshold values indicating completion of grinding; and

actuating discharge flap to discharge the grinded food materials for use and stopping motor (210).

FOR VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI



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Regd. Patent Agent [INPA-1325]

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ABSTRACT

AUTOMATED BI-DIRECTIONAL GRINDING STONE APPARATUS

An apparatus 200 for grinding food materials, the apparatus 200 includes a grinding stone assembly including a stationary grinding base 202 to keep grinding food materials and a grinding roller 204 to grind the food materials; and a moving mechanism comprising a flap linkage 206, a roller linkage 208, an electric motor 210, and a plurality of sensors 212 to control grinding operations such that the grinding roller 204 is rotated over the grinding base 202 including food materials, and the rotational motion of the grinding roller 204 converts into reciprocal motion to cover complete length of the grinding base 202 to grind the food materials in controlled speed of the grinding roller 204 to preserve texture and retain authentic taste of the food products.

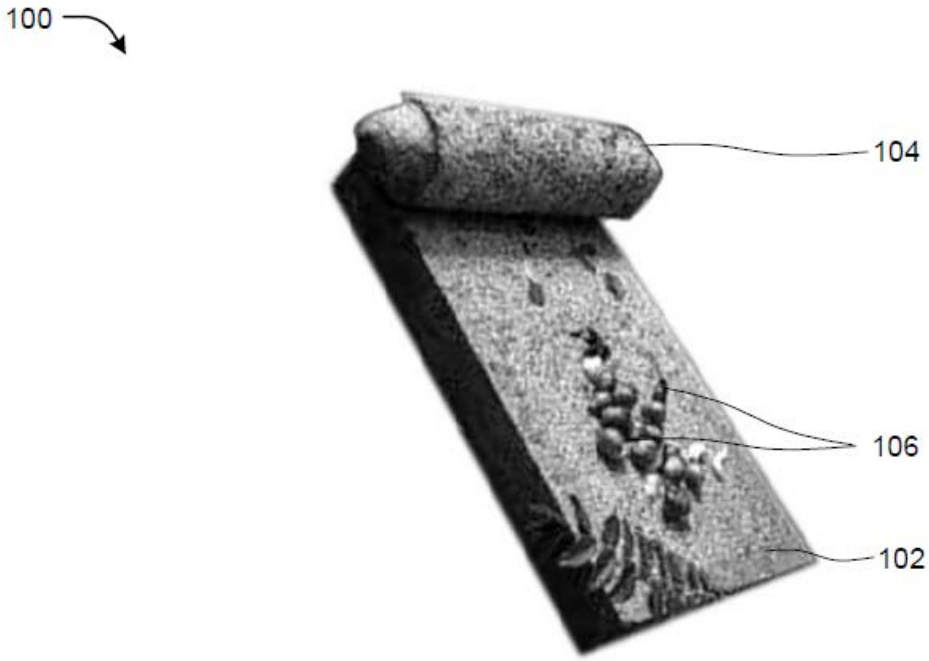


FIG. 2

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